Stochastic Control

Homework Project 1

Reinforcement Learning with Policy Gradient

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Problem 1

a) The state space is $\mathbb{X}$ is defined by the set

$$\mathbb{X} = \{(y, z) | \, y \in \{1, \dots, 5\}, \, z \in \{1, \dots, 10\}\}.$$

b) The action space can be defined as follows

$$\mathbb{U} = \{s, \nu, d\},$$

where, assuming (y, z) is the current state of the the robot,

- s stands for *same*, i.e. the robot stays on the same y and moves only on the z direction with $z \leq 9$,
- ν stands for *up*, i.e. the robot increments both its y and z coordinates with $z \leq 9$ and $y \leq 4$,
- d stands for *down*, i.e. the robot decrements its y coordinate and increments the z coordinate with $z \leq 9$ and $y \geq 2$.

c) Let

$$\mathcal{B} = \{-r_{\text{obst}}, -r_{\text{diag}}, r_{\text{end}}\},$$

where $r_{\text{end}} \gg r_{\text{obst}} > r_{\text{diag}}$. The reward set $\mathcal{R}$ can be defined as follows

$$\mathcal{R} = \left\{ \sum_{x \in \mathcal{A}} x \, \middle| \, \mathcal{A} \subseteq \mathcal{B} \right\}.$$

Then, $\mathcal{R}$ contains the sums of the elements of each subset of $\mathcal{B}$. Note that $0 \in \mathcal{R}$, as the sum of the elements of the empty subset. In this setting award

- $-r_{\text{obst}}$ when the robot steps on an obstacle cell,
- $-r_{\text{diag}}$ when the robot makes a diagonal movement, i.e. the action u is ν or d,
- r_{end} when the robot reaches a desired end position.

If more than one of the above conditions apply for a movement of the robot, then award the sum of the corresponding values. If none apply, then award 0.

d) Let $x = (y, z)$ be the current state and $x^+ = (y', z + 1)$ the next state. Then, once the action u is known both the state transition and the reward are deterministic, i.e.

$$\mathbf{q}(\{x^+, r\}; x, u) = 1,$$

where $r \in \mathcal{R}$ is calculated according to the rules outlined in (c). For example, if $u \in \{\nu, d\}$ and x^+ is a desired end position, then $r = r_{\text{end}} - r_{\text{diag}}$, and if $u = s$ and x^+ is an obstacle-free non-end position, then $r = 0$.
For the state transitions, if

- $u = s$, then $y' = y$ with $z \leq 9$,
- $u = \nu$, then $y' = y + 1$ with $z \leq 9$ and $y \leq 4$,
- $u = d$, then $y' = y - 1$ with $z \leq 9$ and $y \geq 2$.

Note that if $z = 10$, the robot has reached an end position and the path ends. Also note that illegal actions ($u = \nu$ and $u = d$ when $y = 5$ and $y = 1$ respectively) have been taken into account by forcing $\pi(u; x) = 0$.

Problem 2

I am seriously conflicted as to what I'm supposed to write here.
a) The SPG is essentially a special case of the basic PG with $N = 1$. Indeed, basic PG tries out different trajectories per iteration of M and averages them to get the gradient of the objective function $J(\theta)$ and subsequently a better parametrization θ through gradient ascent. The time complexity is $\mathcal{O}(MN)$.
On the contrary, SPG samples one trajectory per update step leading to a more computationally efficient algorithm and computes the next step of the parametrization θ accordingly. Of course, this means that the algorithm may produce inconsistent results if the variance gradient of the objective functions is unbounded. The time complexity is $\mathcal{O}(M)$. Hence, SPG maybe used when time and computational resources are limited.
SPG could also be useful when the actual global optimal parametrization lies very far away from the initial one and hence the updates need to be more drastic in order to avoid other local optima. In this scenario, PG would not be suitalbe given its more conservative update loop.

b) The motivation to employ the SPG instead of the PG stems from the fact that as long as θ updates to the direction of individual paths which on average produce the real gradient of the objective function, then it should work. Indeed, we view θ_k where $k = 1, \dots, M$ as a series of random variables in the same way as section **3.1** of the SGD notes.
Now, let $\omega = (x_0, u_0, \dots, x_T, r_T)$ be a path generated by the policy π^{θ_k}. Then, in similar fashion to the notes, we may write

$$\theta_{k+1} = \theta_k + \eta \nabla_\theta J_\omega(\theta_k),$$

where

$$\nabla_\theta J_\omega(\theta_i) = \sum_{k=1}^{T} R_k \sum_{\ell=0}^{T-1} \nabla_\theta \log \pi^{\theta_i}(U_\ell; X_\ell).$$

But then

$$\mathbf{E}_\omega \left[\nabla_\theta J_\omega(\theta)\right] = \nabla_\theta J(\theta)$$

which makes the estimator of the gradient unbiased as desired. (I have barely convinced myself for all the above...)

c) The 'may be used' part should be answered in part (a). For convergence guarantee, we need the following:

- unbiased estimation, which is self-fulfilled based on (b),
- bounded variance of $\nabla_\theta J_\omega(\theta)$, so that it doesn't diverge or oscillate, and
- a step size η which is neither too small so that the algorithm can make meaningful updates in terms size during the first iterations and nor too big so that it doesn't overtake optimal solutions.

Problem 3